

Automated Discovery of Coupled-Mode Setups

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In optics and photonics, a small number of building blocks—like resonators, waveguides, arbitrary couplings, and parametric interactions—allow the design of a broad variety of devices and functionalities, distinguished by their scattering properties. These devices include transducers, amplifiers, and nonreciprocal devices, like isolators or circulators. Usually, the design of such a system is handcrafted by an experienced scientist in a time-consuming process, where it remains uncertain whether the simplest possibility has indeed been found. In our work, we develop the discovery algorithm AUTOSCATTER, which automates this challenge. By optimizing the continuous and discrete system properties, our automated search identifies the minimal resources required to realize the requested scattering behavior. In the spirit of artificial scientific discovery, it produces a complete list of interpretable solutions and leads to generalizable insights, as we illustrate in several examples. AUTOSCATTER opens the door towards the automated discovery of scattering setups for photonics, microwaves, and optomechanics, with possible future extensions to periodic structures, sensing, and electronic devices.

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I. INTRODUCTION

Waves represent one of the most basic physical phenomena, which explains the importance of applications enabled by wave transport in suitably designed structures, ranging from integrated photonics to microwave circuits. A large class of on-chip devices—from isolators and circulators for nonreciprocal transport [1–4] to hybrid frequency converters [5,6] and new classes of amplifiers [7,8]—can be engineered by combining the right building blocks in a well-chosen scattering topology. So far, each new design in this domain has been proposed based on human ingenuity and experience. It would be desirable to speed up the pace of this laborious exploration process in the high-dimensional space of possible setups and obtain a complete overview of all conceptually distinct options for any given desired functionality. Here, we present the automated discovery algorithm AUTOSCATTER [9], which achieves these goals and also helps to uncover new insights.

Tackling challenges of this kind is the domain of artificial scientific discovery [10]. This rapidly evolving field has the ambitious aim to automate all aspects of the

scientific process using tools of machine learning and artificial intelligence, with an emphasis on interpretability and discovery of new conceptual insights. Pioneering examples are the identification of organic molecules from mass spectrometry by the computer program DENDRAL [11] and the generation and experimental testing of hypotheses in biochemistry by a robot scientist [12]. Important challenges in this field include automatic extraction of symbolic expressions [13,14], discovery of collective coordinates [15,16], automated planning and execution of experiments [17–19], and generating new experimental setups. The latter task has been successfully demonstrated for the preparation of entangled states in quantum optics [20,21], and the optimization of superconducting circuits [22], gravitational-wave detectors [23], and signal processing devices using optical components [24].

Automated discovery relies crucially on casting the complex search space into a unifying language. For the domain envisaged here, we identify coupled-mode theory [25,26] as providing a sweet spot on the continuum between more microscopic hardware-specific descriptions (such as engineered refractive-index distributions [27–30]) and descriptions based on macroscopic building blocks on the level of entire functional elements (like amplifiers). The latter require a larger number of building blocks and need more human design choices; for example, Ref. [24] focuses on signal-processing setups that can be decomposed into networks of specific technical components. In contrast, coupled-mode designs are transferable between different platforms and can always be translated back into several

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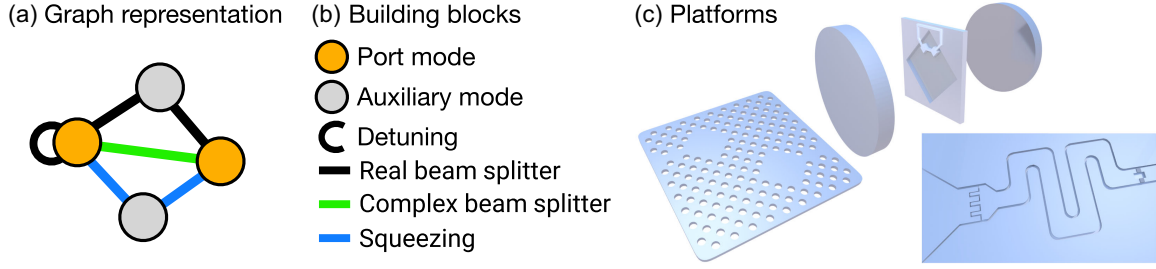


FIG. 1. Graph representation of coupled-mode systems. (a) Physical modes correspond to graph nodes, couplings to edges. Port modes (orange) are connected to waveguides that relay input and output signals. Auxiliary modes (gray) have an intrinsic loss channel whose output signal is not monitored. The edges’ weights are dimensionless coupling rates that are rescaled over the loss rates. (b) Discrete building blocks available to the discovery algorithm. (c) Illustration of some of the many hardware platforms that are efficiently described by coupled-mode theory. These platforms include photonic circuits, e.g., based on photonic crystals (left), optomechanical setups (top), and microwave circuit setups (right).

different concrete physical structures, ranging from photonic crystals to optomechanical setups and microwave circuits. Of equal importance is the formalization of the goal. Here, we choose a description in terms of the scattering matrix, specifying the desired functionality via the externally visible characteristics.

Our discovery algorithm AUTOSCATTER is based on an efficient search process that exploits the recursive modular structure of the problem to prune the exponentially large space of possibilities and provides a complete list of setups fulfilling specified constraints. It automatically ends up with interpretable and generalizable results both by using discrete building blocks and by suggesting only dimensionless parameters of universal, implementation-independent meaning. The final outcome is a menu of irreducible setups that allows one to select optimality criteria even afterwards. Moreover, we introduce a fruitful general extension of the toolbox available in artificial discovery, observing that conceptual insights in design are often based on thinking about idealized structures realizing asymptotic limits of parameters. We describe a modification of our algorithm that exploits this metaprinciple.

We illustrate the benefits of AUTOSCATTER for a variety of target functionalities, including couplers, circulators, and directional amplifiers, where it produces new discoveries and generalizable insights.

II. COUPLED-MODE THEORY

We consider a network of coupled bosonic modes, which can be implemented as an optical, electric, mechanical, or hybrid setup, comprising different types of modes. This device is probed with multiple input signals, and its linear response is encoded in its scattering matrix S . The element S_{ji} is the transmission coefficient of a signal from the input port i to the output port j . In our work, we aim to automatically discover the simplest setups that fulfill a certain target response, which is encoded in a target scattering matrix S^{target} .

The precise characterization of a specific physical implementation of an ideal target device requires a carefully crafted system-specific model. In contrast, the ideal behavior is typically captured by a simple and general high-level description. We choose our artificial scientist to adopt such a high-level description, as a human scientist would do. This approach allows us to discover ideal and transferable solutions. In Section IX, we discuss how to tailor our discovered solutions to specific hardware platforms.

At the highest level of description, any multimode circuit can be represented as a graph with colored edges [31] [see Figs. 1(a) and 1(b)], regardless of the underlying hardware platform. Nodes correspond to modes, and edges represent resonant interactions between the modes. “Port modes” (orange) are connected to waveguides transmitting the input and output signals of interest. The carrier frequencies $\omega_{L,j}$ of the input fields should be close to the respective mode resonances ω_j and can be different for different port modes; e.g., an optical and a mechanical mode will have very different frequencies. Here, we focus on the response of each port mode at the respective carrier frequency, which is described by a single scattering matrix encoding the ideal behavior of the device. AUTOSCATTER could be extended to also consider aspects like optimizing the bandwidth [32], but this would lead to less transferable solutions (see Sec. IX) and is therefore beyond the scope of this work.

The port modes are often insufficient to implement a given scattering matrix. Therefore, we permit one to add a variable number of auxiliary modes (gray). The auxiliary modes are also damped, but we do not monitor the corresponding loss channels. Therefore, these channels are not included in the target scattering matrix.

We use colored edges to distinguish between different types of interactions. The edges’ weights are appropriately rescaled dimensionless coupling constants. We choose the rescaling such that the weights fully determine the scattering matrix and, thus, also incorporate the essential information about the relative strength of the different coherent and dissipative processes. This rescaling is an essential step

in our method and merits a separate discussion below. First, we discuss the coherent interactions between the modes of the multimode device.

The response of state-of-the-art multimode devices can be precisely characterized by linearizing the full nonlinear dissipative dynamics about a steady state. In this setting, the interactions between N high-quality-factor modes are well described by a time-independent bosonic quadratic Hamiltonian of the form

$$\frac{1}{2} \sum_{i,j=1}^N (g_{ij} \hat{a}_i^\dagger \hat{a}_j + \nu_{ij} \hat{a}_i^\dagger \hat{a}_j^\dagger) + \text{H.c.} \quad (1)$$

Here, H.c. denotes the Hermitian conjugate, and $\hat{a}_j$ and $\hat{a}_j^\dagger$ are the ladder operators for mode j . Any off-diagonal nonzero coupling constant, $\nu_{ij} \neq 0$ or $g_{ij} \neq 0$, corresponds to a graph edge. Importantly, there are two distinct types of interactions: Squeezing interactions with coupling constants ν_{ij} (corresponding to blue edges in our graph representation) create pairs of excitations in modes i and j , while beam-splitter interactions with coupling constants g_{ij} exchange excitations between modes i and j . In our optimization, we also distinguish between complex-valued beam-splitter couplings (green) with $g_{ij} \in \mathbb{C}$, which involve a complex phase depending on the direction of the exchange, and real-valued beam-splitter couplings (black) with $g_{ij} \in \mathbb{R}$, which are direction independent. It is useful to distinguish between the two different types of beam-splitter interactions in view of identifying couplings that can be implemented passively in many common experimental scenarios (see Sec. IX for more details).

An important aspect of the Hamiltonian in Eq. (1) is that it is defined in a rotating frame. By applying a so-called “rotating-wave approximation” (RWA), we drop the fast oscillating terms describing nonresonant interactions (see Appendix D for more details). In this rotating frame, each mode j rotates at its corresponding carrier frequency $\omega_{L,j}$ (for an auxiliary mode j , $\omega_{L,j}$ is the vibration frequency induced by a signal entering the device at any port l with the respective carrier frequency $\omega_{L,l}$). Thus, $-g_{jj}$ is interpreted as the detuning Δ_j of the carrier frequency $\omega_{L,j}$ from the mode resonance ω_j , $\Delta_j = \omega_{L,j} - \omega_j$. In our graph representation, any nonzero detuning is visualized as self-loop [see Fig. 1(a)]. Importantly, the RWA Hamiltonian already captures the ideal behavior of the devices even though it does not directly depend on the resonance frequencies ω_j . This appealing feature makes any insights gained using this approximation highly transferable across various platforms. For this reason, it is widely adopted for the handcrafted design of multimode circuits [2–6,8,31–44]. A more refined description including nonresonant interactions is often used to quantify how a specific implementation deviates from the ideal target behavior. This deviation is typically small for high-quality

factor modes ubiquitous in modern quantum optics platforms. Moreover, it is device specific; thus, incorporating it in our approach would lead to less transferable results.

As anticipated above, a key novelty of our graph representation is our rescaling of the weights. Previous works adopting graph representations of multimode circuits [31,32] introduced weights by expressing the coupling constants and other frequency parameters in units of a single base frequency. This approach is also the most common in other numerical and analytical studies of the scattering matrix. Here, the number of free parameters is reduced by 1 without loss of generality. Instead, we achieve a much larger reduction of the free parameters by incorporating in the graph edges N decay rates (one for each mode). We first express the Hamiltonian in Eq. (1) in matrix form, i.e., in the first quantization, as a so-called Bogoliubov–de-Gennes (BdG) Hamiltonian. Then, we rescale the BdG Hamiltonian in the following way (see Appendix C):

$$H = \frac{1}{\sqrt{\kappa}} \begin{pmatrix} g & \nu \\ \nu^* & g^* \end{pmatrix} \frac{1}{\sqrt{\kappa}}. \quad (2)$$

Here, κ is the diagonal matrix $\kappa = \text{diag}(\kappa_1, \dots, \kappa_N, \kappa_1, \dots, \kappa_N)$, and κ_i is the decay rate of mode i (out-coupling rate for port modes, intrinsic loss rate for auxiliary modes). Our graph weights are then the entries of this dimensionless BdG Hamiltonian. Importantly, this approach does not involve any loss of generality since, in our high-level description based on the RWA, the scattering matrix is fully determined by the dimensionless BdG Hamiltonian H (see Appendix F).

The importance of the rescaling in Eq. (2) is twofold. On one hand, the weights H_{ij} acquire a universal implementation-independent meaning as they are now directly related to the dimensionless physical parameters that determine the scattering behavior, i.e., the dimensionless detuning $(\omega_{L,i} - \omega_i)/\kappa_i = -H_{ii}$, the cooperativities $C_{ij} = 4|H_{ij}|^2$, and the gauge-invariant phases accumulated on closed loops (synthetic field fluxes), $\Phi_{i,j,l,\dots,k} = \arg(H_{ij}H_{jl}\dots H_{ki})$. On the other hand, it implies that the algorithm discovers classes of solutions that all obey the desired target scattering behavior. When translating back to a concrete physical setup, we are left with the freedom to arbitrarily choose the decay rates κ_i and, thus, fix the couplings accordingly (although, for amplifiers, some additional stability constraints might apply, see Appendix G).

For concreteness, we focus below on the important special case of phase-preserving devices. In this setting, the equations of motion for $(\hat{a}_1, \dots, \hat{a}_N, \hat{a}_1^\dagger, \dots, \hat{a}_N^\dagger)$ break down into two decoupled sets of equations, which are related to each other via the particle-hole symmetry. As a result, the scattering matrix is an $N \times N$ matrix. For phase-preserving amplifiers, we enforce the block structure of the equations of motion by following the common engineering

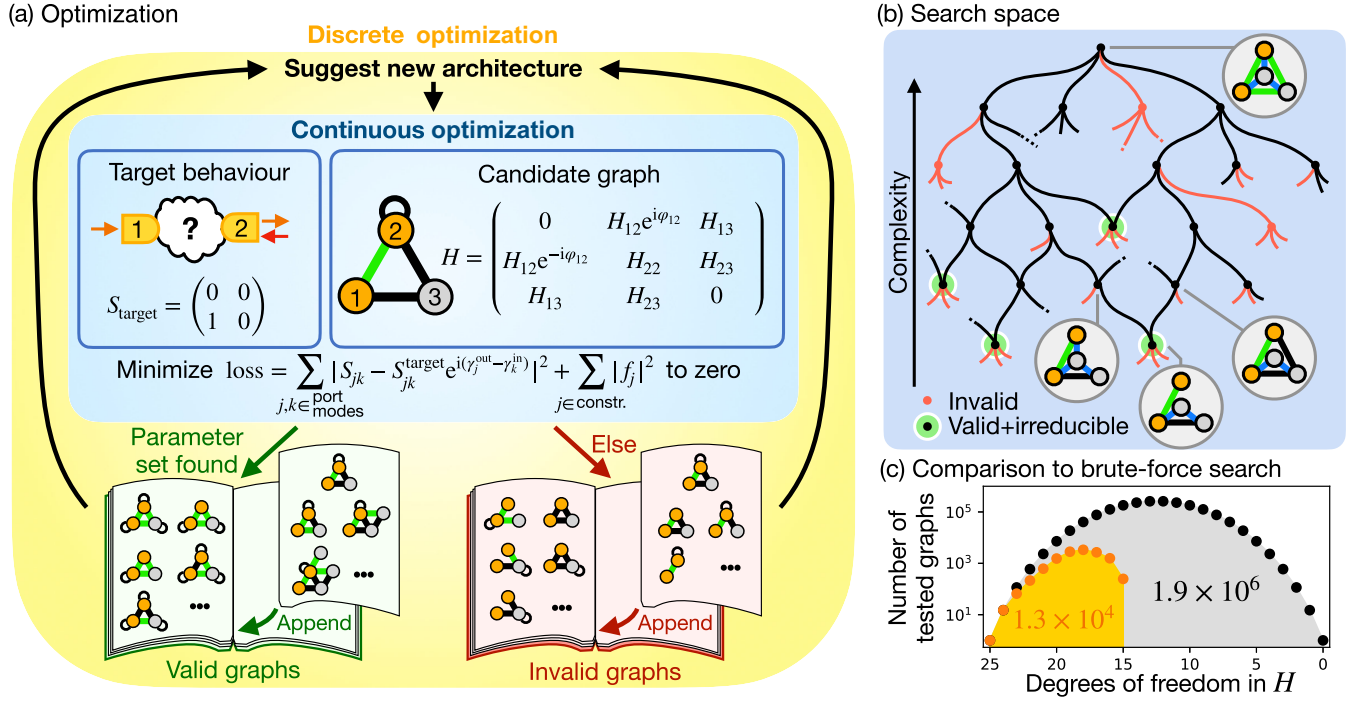


FIG. 2. Automated discovery of modular scattering setups. (a) Optimization scheme behind AUTOSCATTER. The discrete optimization sorts graphs into the libraries of valid (green book) and invalid (red book) graphs. In every round, a new graph is suggested and, depending on the outcome of the continuous optimization, sorted with all its extensions or subgraphs into the library of valid or invalid graphs. The continuous optimization minimizes the loss function to realize a target scattering behavior (left block), which is an isolator here. The candidate graph (right block) defines the number of auxiliary modes and an allowed set of nonzero coupling rates and phases in the dimensionless Hamiltonian H . (b) Search space (schematic). The search starts from the smallest fully connected graph realizing the target behavior (topmost graph). From there, our method successively prunes edges, resulting in multiple irreducible solutions. (c) Comparison of the number of tested graphs for the directional coupler [see Fig. 4(b)], which uses five modes. The total number of possible graphs (black) for a certain number of free variables in H is compared to the number of graphs that actually had to be tested by our exhaustive search. Our scheme starts with maximally connected graphs (left end) and successively goes to more restricted graphs (right end). By testing only around 13,000 graphs, we are able to characterize all 1.9 million graphs in this example.

principle of dividing the modes (graph nodes) into two disjoint sets [2,31,33,37,39]. The modes within the same set are then only coupled via beam-splitter couplings, while the modes in different sets are only coupled via squeezing interactions. In this way, the annihilation operators for the modes in one set are only coupled to the creation operators of the modes in the other set (see Appendix F for more details). This constraint restricts the set of graphs considered in our optimization. For example, a triangle with just one blue edge will be excluded. We do not indicate explicitly which mode belongs to the same group because this can be easily deduced from the colors of the edges (see Appendix F).

III. OPTIMIZATION SCHEME

To identify setups that implement the target scattering matrix, AUTOSCATTER has to explore the discrete search space of all possible graphs and, given a graph, find an appropriate set of values for the underlying continuous parameters, i.e., the nonzero cooperativities and coupling phases. To this end, AUTOSCATTER performs a two-step

procedure: A discrete optimization routine suggests new graphs, and an embedded continuous optimization algorithm looks for appropriate values of the parameters [see Fig. 2(a)]. If the continuous optimization is successful, we label a graph as valid; i.e., this graph fulfills the desired scattering behavior.

First, we discuss the discrete optimization [see Fig. 2(a)]. We aim to identify the simplest valid graphs as they represent the simplest experimental implementations of the target response. Often, there are a multitude of valid graphs that differ in the number of auxiliary modes and the number and complexity of the required couplings. In a first step, AUTOSCATTER tests only fully connected graphs for validity, starting from the graph with zero auxiliary modes. It increments the number of auxiliary modes until it finds the minimum number of auxiliary modes required to achieve the target behavior. This graph is the smallest fully connected graph that satisfies the target behavior. Then, AUTOSCATTER prunes this fully connected graph and identifies “irreducible” graphs that cannot be further simplified by setting any coupling rate or phase to zero.

To identify all such graphs, we perform a complete enumeration.

Because of the large discrete search space, a brute-force approach testing all graphs is infeasible in many cases of interest. We solve this problem by leveraging the knowledge gained from previously tested graphs, drastically reducing the required computational resources. We note that an alternative approach could be developed based purely on continuous optimization, introducing terms in a loss function that reward sparsity (zero couplings), as in Refs. [45–47]; however, this approach would not enable us to obtain a complete set of solutions.

We take advantage of the fact that removing an edge from a graph is equivalent to setting a coupling rate to zero. Thus, if a graph is found to be invalid, it carries over to all graphs, which can be generated by removing some edges and/or setting coupling phases to zero. Likewise, all extensions of a valid graph are valid [see Fig. 2(b)]. This observation allows us to efficiently construct exhaustive libraries of valid and invalid graphs by testing only a small fraction [see Fig. 2(c)]. In Sec. VIII, we give an in-depth analysis of the search-space reduction provided by AUTOSCATTER. See also Appendix A for further details on the discrete optimization.

We now turn to the continuous optimization. The scattering matrix S is fully determined by the dimensionless Hamiltonian H defined in Eq. (2), which contains the free coupling parameters. We look for setups implementing the desired scattering behavior by minimizing the squared deviation loss function $\mathcal{L}$ with respect to those couplings:

$$\mathcal{L} = \sum_{j,k \in \text{port modes}} |S_{jk} - S_{jk}^{\text{target}} e^{i(\gamma_j^{\text{out}} - \gamma_k^{\text{in}})}|^2 + \sum_{j \in \text{constr}} |f_j|^2. \quad (3)$$

We note that the target scattering matrix S^{target} can contain further free parameters that are not specified by the desired scattering properties. For example, we might impose that two scattering amplitudes are equal, but a suitable value needs to be discovered. In addition, we introduce the free gauge parameters $\gamma_i^{\text{in/out}}$. These parameters reflect the freedom of choosing independently, for each port, a reference quadrature and a reference position where the fields are measured. For more details, including how to fix the gauge in specific experimental settings, see Appendix J. Finally, we allow for extra constraints $f_j = 0$ that can be used, e.g., to enforce minimum added noise as detailed in Appendix H. We minimize the loss $\mathcal{L}$ as a function of the free parameters in H , S^{target} , and the phases $\gamma_i^{\text{in/out}}$ (see Appendix B for more details). A valid solution is obtained whenever the loss reaches zero.

The optimization described within this section is the main part of AUTOSCATTER. In addition, AUTOSCATTER provides an automated analysis of the complexity of the discovered solutions, as discussed in Sec. IX. The broader

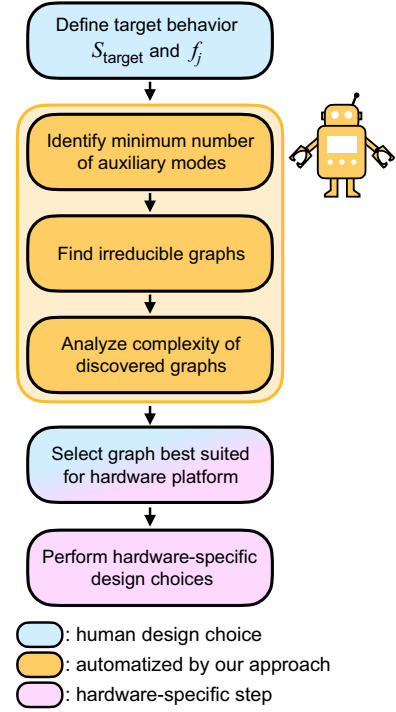


FIG. 3. Flowchart showing the full design pipeline of coupled-mode systems. The steps automatized by AUTOSCATTER are highlighted in orange. The remaining steps, not covered by our method, are hardware specific (see Sec. IX).

design pipeline for a specific experimental platform, also including steps not covered by AUTOSCATTER, is shown in the flowchart in Fig. 3.

In the following, through a series of illustrative examples, we demonstrate how AUTOSCATTER can design scattering setups, some of which have been overlooked by handcrafted methods. We also show that our approach can automate several key aspects of scientific discovery, which include the generalization of design patterns, the finding of explicit analytical expressions for continuous classes of problems, and the discovery of asymptotic solutions.

IV. SIMPLE EXAMPLE: ISOLATOR

As a first proof of concept, we apply AUTOSCATTER to implement an isolator, i.e., a two-port device with perfect transmission from the input to the output and zero reverse transmission [see Fig. 4(a)]. Our method shows that the simplest solution has one auxiliary mode and a single complex-valued beam-splitter coupling. The solution has the cooperativities $C_{12} = C_{23} = C_{31} = 1$ and the synthetic flux $\Phi_{123} = \pi/2$. In this case, the unique irreducible solution was already discovered with handcrafted approaches, as in Refs. [2,31,33–35].

Since our solutions are valid for arbitrary decay rates, they also cover the special limiting case in which the auxiliary modes are quickly decaying and act as Markovian baths, mediating dissipative interactions [36].

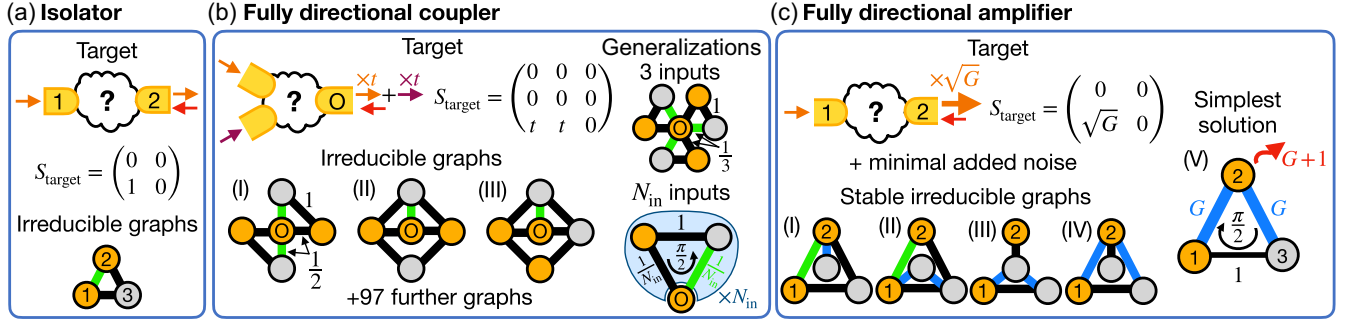


FIG. 4. Illustrative examples for automated discovery of scattering setups. Target scattering behaviors and the corresponding irreducible graphs identified by AUTOSCATTER are shown. (a) Isolator. Only the signal from mode 1 to 2 is transmitted. (b) Directional coupler. Multiple input signals are summed up and transmitted to the output port. Graphs (I)–(III) are the smallest irreducible graphs for a directional coupler with two inputs. Graph (I) has the smallest number of couplings; graphs (II) and (III) have the smallest number of parametric drives. The other identified 97 further irreducible graphs (not shown) require either more resources or are gauge transformations of the graphs shown. Graph (I) can be generalized to more inputs (right panel). To realize a coupler with N_{in} inputs, one has to repeat the graph structure (bottom right, blue background) N_{in} times. Analytical terms discovered for the cooperativities and fluxes are indicated; the transmission turns out to be $t = 1/\sqrt{N_{\text{in}}}$. (c) Directional amplifier. Input from port 1 is amplified by a factor of $\sqrt{G}$; the other direction is blocked. The device has minimal backaction, and the noise is quantum limited. Graphs (I)–(IV) are the simplest solutions without making use of intrinsic loss on the port modes. Allowing intrinsic losses, graph (V) is the simplest graph. Analytical terms discovered for the cooperativities C_{ij} , the intrinsic loss rescaled by the out-coupling (red arrow), and the flux Φ_{132} are indicated. All graphs shown are stable. The unstable solutions are further discussed in the Supplemental Material [48].

In Supplemental Material [48], we provide a mapping from the graph’s weights to the Lindblad master equation of a reservoir engineering scheme. Returning to the specific isolator example, this mapping allows us to recover the dissipative isolator discussed in Ref. [36].

In addition to finding the simplest solution, any valid reducible graph provided by our method can be used as an ansatz by an experienced researcher to find the analytical expression for the corresponding manifold of solutions. The numerical data can provide further insights to refine the initial ansatz, e.g., suggesting constraints in the graph parameters. As an example of this approach, we find the analytical expressions that describe (up to gauge transformations) all three-mode solutions of the isolator problem and generalize the solution presented in Refs. [2,31,33–36]. By using the empirically observed constraints that all dimensionless detunings and all cooperativities assume the same values, we find a single analytical expression parametrized by the flux Φ_{123} ,

$$C_{ij} = 1 + 4 \left(\frac{\Delta_i}{\kappa_i} \right)^2, \quad \frac{\Delta_i}{\kappa_i} = \frac{1}{2} \tan(\Phi_{123} - \pi/2), \quad (4)$$

for $j \neq i$, and $i, j = 1, 2, 3$. For later discussion, we note that, after replacing the auxiliary mode with a third port mode, the same solution above describes the most general three-mode circulator.

V. DISCOVERING GENERALIZABLE DESIGN PATTERNS

An important goal of scientific discovery is to create insights that can be used to derive more general solutions.

A common example of such an insight is a design pattern discovered in solving a simple problem but that can also be reused as a building block to solve more complex problems. Graphical representations inherently can be leveraged to gain insights. For example, chemical graphs are used by DENDRAL to interpret mass spectra [11]. In addition, the authors of Refs. [20,45–47] discovered a new graph-based scheme to create many-body entangled quantum states, allowing them to automatically discover graph solutions with a small number of particles and to generalize them to more particles.

In our setting, we demonstrate how our graph-based solutions can be generalized to realize devices with an arbitrary number of ports. This example shows that the weights in the solutions of AUTOSCATTER are typically unique or in a low-dimensional manifold, contributing to their interpretability. This appealing feature is achieved by focusing on irreducible graphs and by our rescaling in Eq. (2). It is also the reason why we have been able to derive analytical expressions from most of the numerical solutions provided by AUTOSCATTER (see the following example and Secs. VI and VII).

A. Example: Fully directional coupler

As an example, we consider a fully directional coupler. This device combines signals from multiple input ports and transmits them with equal transmission amplitude to an output port. Here, we aim for a fully directional coupler with zero reflection and reverse-transmission amplitudes [see Fig. 4(b)]. We leave the transmission amplitude t as a free parameter. Our goal is to find a general solution valid for an arbitrary number of input ports N_{in} .

First, we use AUTOSCATTER to find possible solutions for $N_{\text{in}} = 2$. It turns out that at least two auxiliary modes are required to realize a two-port device. In total, we find 100 different irreducible graphs that differ in the number and complexity of the underlying couplings. We select the graphs with the minimal number of couplings, graph (I), and the graphs that require the smallest number of parametric pumps, graphs (II) and (III). For an explanation of how to count the number of pumps, see Sec. IX. The remaining irreducible graphs are either gauge transformations of these graphs or they require more resources in terms of couplings and parametric pumps.

Inspecting graph (I), we note that it can be viewed as the combination of two isolator graphs, which inspires a general ansatz for an N_{in} -port coupler: a graph that comprises an isolator building block for each input port [see the right panel in Fig. 4(b)]. We then use our continuous optimization routine to verify the validity of the ansatz and calculate the free parameters for the first few N_{in} , up to $N_{\text{in}} = 5$, corresponding to $N = 11$ nodes. For such large graphs, an exhaustive search would be daunting. From the solutions, we infer the weights for arbitrary N_{in} : All cooperativities corresponding to the interaction between an input mode and its respective auxiliary mode take the same value, $C_{\text{in,aux}} = 1$; the cooperativities corresponding to the coupling between the output mode and other modes take the value $C_{\text{out,in/aux}} = 1/N_{\text{in}}$. The synthetic field fluxes are the same for all isolatorlike loops, $\Phi_{\text{in,out,aux}} = \pi/2$. Finally, the transmission turns out to be $t = 1/\sqrt{N_{\text{in}}}$, implying that no noise from the auxiliary modes is injected into the output port.

VI. SOLVING A CONTINUOUS CLASS OF PROBLEMS

Many interesting physical problems are parametrized by continuous parameters, e.g., the temperature or the value of an external field. Human researchers typically attempt to tackle these problems by deriving analytical solutions that are naturally interpretable. This approach becomes even more interesting when applied to optimization problems where the objective is continuous, e.g., when optimizing a variational quantum algorithm [53] to realize a continuous set of quantum gates.

In our multimode circuit setting, one is often interested in a class of problems parametrized by one or more continuous parameters in the target scattering matrix. This type of challenge can be solved using our algorithm by first identifying a valid graph using a fixed set of values of the parameters. Then, we use the continuous optimization to create a dataset of solutions for varying parameter values (but fixed graph structure). Finally, we attempt to derive closed analytical formulas for the cooperativities and fluxes by symbolic regression. This scheme can also be used in similar optimization problems with continuous objectives as discussed above.

A. Example: Directional quantum-limited amplifier

For illustration purposes, we consider several recent works [2,31,33,36–39] that have proposed schemes to realize on-chip directional quantum-limited amplifiers using multimode circuits to replace bulky state-of-the-art ferrite-based devices. The target device has arbitrary gain G and is fully directional, i.e., has zero backreflection, and zero reverse transmission. Moreover, both at the input and the output ports, the added noise should reach the so-called quantum limit, i.e., the fundamental limit set by the laws of quantum mechanics [40]. These requirements can be translated into constraints f_j [see Eq. (3)] on the scattering matrix S that depend on the temperatures of the input and output ports (see Appendix H for more details). For both input and output ports at zero temperature, AUTOSCATTER recovers the solution used in Refs. [2,31,33,36] (not shown). In this solution, the added noise in the output port is injected into the multimode system from the same port. For this reason, the device will not be quantum limited if the output port is hot.

An interesting open problem in this context is to find the simplest design for a fully directional amplifier that is still quantum limited in the presence of a hot output port [39] [see Fig. 4(c)]. Liu *et al.* [39] addressed this problem with a handcrafted approach, discovering two small architectures comprising two auxiliary modes and six couplings. However, applying AUTOSCATTER, we discovered several architectures [see graphs (I)–(IV) in Fig. 4(c)], which require less couplings and can also be implemented with less parametric pumps (see Sec. IX).

In our description in Sec. II, we assumed, for simplicity, that the port modes' only loss channel is the out-coupling to some waveguide. In general, port modes can be subjected to further losses, e.g., an intrinsic loss channel, which is not monitored. Using the loss rates of these additional loss channels as degrees of freedom in the continuous optimization (see Appendix F for more details), AUTOSCATTER finds graph (V) in Fig. 4(c). This solution comprises a single auxiliary mode and three couplings, and has an undercoupled output port; i.e., most of the signal is dissipated into a (zero-temperature) loss channel. While this setup is not power efficient, it fulfills all the imposed goals and represents an instructive counterintuitive solution since one could naively expect that the loss of a large part of the signal would be incompatible with quantum-limited amplification.

To discover the graphs in Fig. 4(c), we run the optimization with a constant value for the gain factor G . In a second step, we use our continuous optimization to create a dataset of solutions for different values of G while keeping the graph structure fixed. Using simple symbolic regression, we are able to derive closed analytical formulas for the cooperativities and fluxes, as well as the full scattering matrix for all discovered irreducible graphs [see Supplemental Material [48] and Fig. 4(c)].

VII. DISCOVERING ASYMPTOTIC SOLUTIONS

In many physical systems, an ideal target behavior can be reached only in the special limiting case of one or more system parameters that tend to infinity or zero, e.g., a strong coupling limit. Even though this ideal limit is strictly unattainable in practice, adopting it promotes simplification and conceptual understanding. How can we introduce such fruitful asymptotic analysis in the context of artificial scientific discovery? One key complication is that multiple parameters often have to assume ideal limits simultaneously and that the way these limits are approached determines the desired target behavior.

We first discuss a simple scenario featuring only two parameters. Consider the task of applying an ideal gate to a dissipative qubit by driving it with a pulse of appropriate amplitude λ and duration T . During a gradient-based optimization run, the gate time T approaches the asymptotic limit $T \rightarrow 0$ to reduce dissipative effects. At the same time, the amplitude λ goes to infinity to compensate for the shorter pulse. Finally, the asymptotic behavior, e.g., $\lambda \approx \pi/T$ for a π pulse, can be extrapolated from the data [see Fig. 5(a)]. In this approach, one can deduce both the scaling law $\lim_{T \rightarrow 0} \lambda T = \text{const}$ and the numerical value of the constant from the data. In quantum circuits consisting of multiple gates, one now has multiple amplitudes and gate times. Instead of optimizing those gate parameters directly, one can reformulate the optimization task by assuming instantaneous pulses parametrized by the pulse area λT . In this way, one builds the prior knowledge of the scaling law into the model and obtains the respective values for

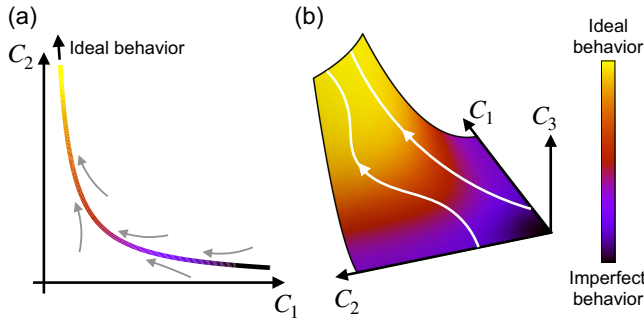


FIG. 5. Asymptotic solutions in automated discovery. (a) Possible asymptotic behavior in a two-dimensional parameter space. The ideal behavior is achieved when C_1 tends to zero and C_2 to infinity while asymptotically reaching a curve. The color represents the loss on this curve, as indicated by the color bar. The gray arrows indicate possible paths in parameter space that a gradient-based optimization could take. In the driven lossy qubit example, the parameter C_1 corresponds to the gate time T , and C_2 to the drive amplitude λ . (b) Asymptotic behavior in higher-dimensional parameter spaces. In this case, the three parameters C_1 , C_2 , and C_3 should diverge while approaching a specific plane to achieve the ideal behavior. The loss on this plane is shown, together with two, out of infinitely many, possible paths of approach to the ideal behavior.

$\lambda T = \text{const}$ directly without requiring extrapolation. This instantaneous pulse concept can be reused as a building block and is naturally used to efficiently solve complex optimization tasks involving more than one qubit and many pulses of different types in a variational quantum circuit [53].

Now, let us consider a more general setting with three or more dimensionless parameters that reach an unattainable limit [see Fig. 5(b)]. As for the qubit problem, we would like to reparametrize it in a way that allows us to take the asymptotic limit and subsequently perform our optimization without requiring extrapolation. For the reparametrization, we need the scaling laws of all parameters in terms of one of the parameters reaching the unattainable limit. However, these scaling laws are *a priori* unknown. Moreover, in a less-constrained problem, the parameters could approach the unattainable limit on trajectories with different asymptotic limits, making it difficult to extrapolate the potentially infinite number of asymptotic solutions or even the scaling laws of the parameters [see Fig. 5(b)].

To address these general challenges, we propose an approach based on asymptotic building blocks. This concept can be applied to a broad range of systems that can be represented as weighted graphs. Just like the instantaneous pulse building block for the previously mentioned example of variational circuits, our building blocks build into the graph model an assumption regarding the scaling laws of the parameters. Here, a building block is a node or group of nodes that can be added to the graph, and it is defined by the scaling laws for the corresponding weights. This approach allows us to combine different building blocks and graph topologies to test a variety of asymptotic ansatzes automatically.

To be more specific, we return to the discovery of scattering setups. In our experience, an asymptotic solution is often required in the presence of experimental constraints that prevent the port modes from interacting directly [5,6,37,41]. This scenario is widespread for hybrid devices, including transducers [5,6], isolators [41], and amplifiers [37]. One or more auxiliary bus modes (for example, mechanical modes in optomechanical setups) are needed to mediate the interactions between port modes. To achieve the ideal behavior, the cooperativities $C_{a_j b_k}$ of the couplings between the bus modes $\hat{b}_k$ and the port modes $\hat{a}_j$ and possibly also the dimensionless detunings of the bus modes $\sqrt{C_{b_k b_k}}$ must diverge. On the other hand, the ratios of cooperativities $C_{a_j b_k}/C_{a_l b_k}$ or $C_{b_k b_k}/C_{a_l b_k}^\alpha$ (with suitable exponents α , depending on the situation) tend to a constant.

We can gain intuition on these asymptotic solutions and incorporate the respective scaling law as a building block in AUTOSCATTER by integrating out one of the modes ($\hat{b}$ or $\hat{a}$ depending on the situation). In this way, we map the original graph to a smaller effective graph whose parameters depend only on the appropriate ratios of cooperativities in the asymptotic limit. This approach allows us to explicitly take the asymptotic limit and later obtain the

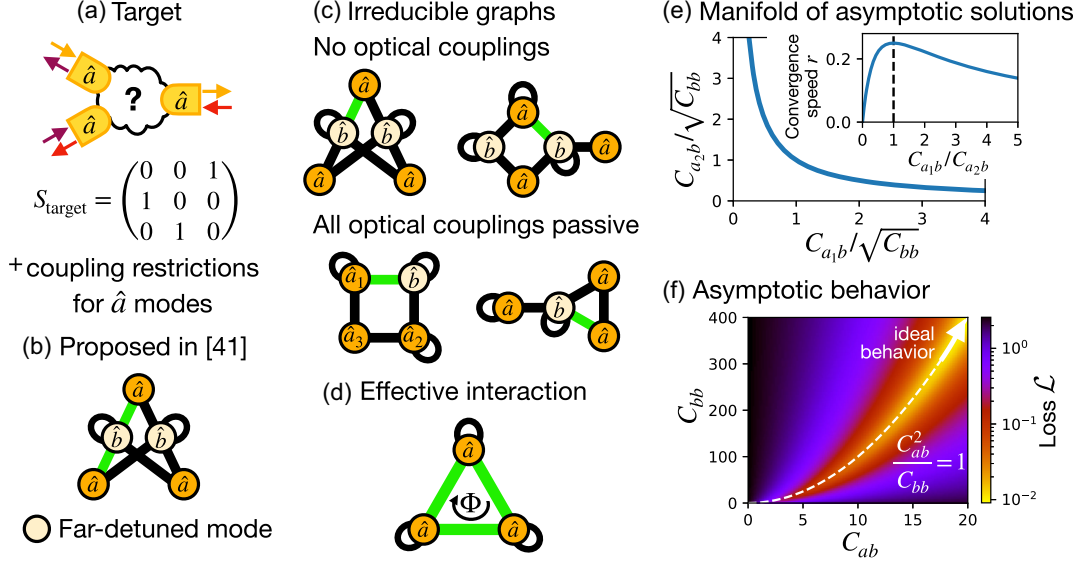


FIG. 6. Optomechanical circulator. (a) Target operation: Realize a circulator between optical modes ($\hat{a}$) by mediating the interaction using mechanical modes ($\hat{b}$). (b) Architecture proposed in Ref. [41] when no direct couplings are allowed between the optical modes. Instead, the interaction is mediated by far-detuned mechanical modes. (c) Irreducible graphs identified by AUTOSCATTER when no direct couplings are allowed between the optical modes (top) or when the couplings between two optical modes must be passive (bottom). The graphs shown have the smallest possible number of couplings. There exist more irreducible graphs that require more couplings (not shown). (d) Effective interaction between the optical modes after integrating out the far-detuned modes in any of our discovered graphs. This graph resembles a generalized form of the well-known circulator scheme discussed in Refs. [2,31,33–35]. Panels (e) and (f) refer to the bottom left graph in panel (c). (e) Manifold of asymptotic solutions. Every point on the curve $(C_{a_1b}/\sqrt{C_{bb}})(C_{a_2b}/\sqrt{C_{bb}}) = 1$ represents an asymptotic solution that approaches the ideal behavior for $C_{bb} \rightarrow \infty$. The inset shows the convergence speed r as a function of the ratio C_{a_1b}/C_{a_2b} of the asymptotic parameters $C_{a_1b}/\sqrt{C_{bb}}$ and $C_{a_2b}/\sqrt{C_{bb}}$. (f) Cross section of the square deviation loss $\mathcal{L}$ between the scattering matrix S and the target matrix S_{target} . We fix $C_{ab} \equiv C_{a_1b} = C_{a_2b}$. Here, $\mathcal{L} \rightarrow 0$ in the asymptotic limit $C_{ab}, C_{bb} \rightarrow \infty$ with $C_{ab}^2/C_{bb} = 1$ (dashed white line).

asymptotic ratios using the continuous optimization of AUTOSCATTER. For example, consider a set of far-detuned auxiliary bus modes b_k as in Ref. [41]. The bus modes are only virtually occupied and, thus, mediate a purely coherent interaction between two or more port modes a_j . In this case, we can integrate them out to obtain an effective graph for the port modes a_j . In the new graph, the effective cooperatives $C_{jl}^{\text{eff}} = C_{a_jb_k}C_{b_ka_l}/C_{b_kb_k}$ depend only on the asymptotic parameters $C_{a_jb_k}/\sqrt{C_{b_kb_k}}$ (see Supplemental Material [48]).

Currently, our approach is limited to using certain asymptotic building blocks, which have to be predefined by a human expert. The same expert-aided approach could be adopted in other artificial scientific discovery problems, e.g., featuring a varying number of masses or charges of very different sizes. An important goal for future research is to automatically identify such limits and automatically define such new concepts.

A. Example: Optomechanical circulator

We apply this extended optimization scheme to discover the simplest setup implementing an optomechanical circulator in the experimental scenario considered in Ref. [41].

Three microwave port modes are coupled via the optomechanical interactions mediated by a varying number of mechanical modes. No direct coupling is available [see Fig. 6(a)]. Since each optomechanical coupling requires a different laser drive, the key measure of experimental complexity in this case is the overall number of couplings. The handcrafted architecture proposed in Ref. [41] requires two far-detuned mechanical modes and six couplings [see Fig. 6(b)]. Our optimization scheme finds a simpler setup, which requires the same number of far-detuned modes but only five couplings [see Fig. 6(c)]. If we apply the softer constraint where passive couplings are allowed between the optical modes, our scheme identifies even simpler graphs, only involving one far-detuned mode [see Fig. 6(c)].

As we explained above, AUTOSCATTER translates the asymptotic graphs with far-detuned modes into an effective graph featuring only port modes. Our effective graphs improve the interpretability of the solutions. For example, all valid asymptotic graphs in our optomechanical circulator are translated into the same effective three-mode circulator graph shown in Fig. 6(d) (see Supplemental Material [48] for details). As we discussed in Sec. IV, this graph encodes the most general three-mode solution for an ideal circulator or isolator [see Eq. (4)]. We can leverage

this solution to discover the manifold of asymptotic parameters corresponding to ideal solutions for our discovered asymptotic graphs. As an example, consider the lower-left graph in Fig. 6(c). Here, all passive couplings between the optical modes would be allowed. However, they cannot realize the required synthetic field flux (see Sec. IX). This role is fulfilled by the mechanical mode, which mediates an effective complex-valued beam-splitter coupling between $\hat{a}_1$ and $\hat{a}_2$. We empirically find that this architecture only has solutions where the effective detuning is zero. Therefore, the effective cooperativity $C_{a_1 a_2}^{\text{eff}}$ should fulfill $C_{a_1 a_2}^{\text{eff}} = 1$ [see Eq. (4)]. Taking into account that $C_{a_1 a_2}^{\text{eff}} = C_{a_1 b} C_{a_2 b} / C_{bb}$, this constraint identifies a manifold of ideal solutions for the asymptotic parameters $C_{a_1 b} / \sqrt{C_{bb}}$ and $C_{a_2 b} / \sqrt{C_{bb}}$, as visualized in Fig. 6(e).

In realistic experimental scenarios, the coupling strength and loss rates have upper and lower bounds limiting how close one can approach the asymptotic limits [see Fig. 6(f)]. For example, for the lower-left graph in Fig. 6(c), the loss function $\mathcal{L}$ scales like $\mathcal{L} \sim r / C_{bb}$ in the asymptotic limit. Here, r is a constant that can be interpreted as a convergence speed. Even though in the asymptotic limit all solutions are ideal, they differ in convergence speed. In our example, when we send the dimensionless detuning C_{bb} to infinity, the convergence speed r depends on the ratio of the cooperativities [see inset in Fig. 6(e)]. For practical applications, similar considerations can be used to identify a preferred path to approach the asymptotic limit.

VIII. ANALYSIS OF THE SEARCH-SPACE REDUCTION

The main achievement of our work is the automated discovery of interpretable and transferable coupled-mode setups. In addition, we have introduced an efficient discrete optimization algorithm that allows us to reduce the search space compared to a brute-force search. An interesting open question is how large the reduction in the search space will be. Considering that our algorithm can be directly applied to solve a large class of problems whose potential solutions can be encoded in graphs with continuous weights, this question is of interest beyond our setting of coupled-mode setups.

In Fig. 7, we show the number of graphs tested by AUTOSCATTER in comparison to the total number of graphs within the search space. Each data point corresponds to a different target device, as identified by the target scattering matrix.

As we can see, the performance gain in comparison to a brute-force approach was highly significant in some of the most complex problems. If we quantify the performance gain as the ratio of the number of graphs in the search space over the number of tested graphs, it varies in a broad range from about 5 [see, e.g., data point (vii)] to around 400 in our experiments [see data point (ix)]. As one might intuitively

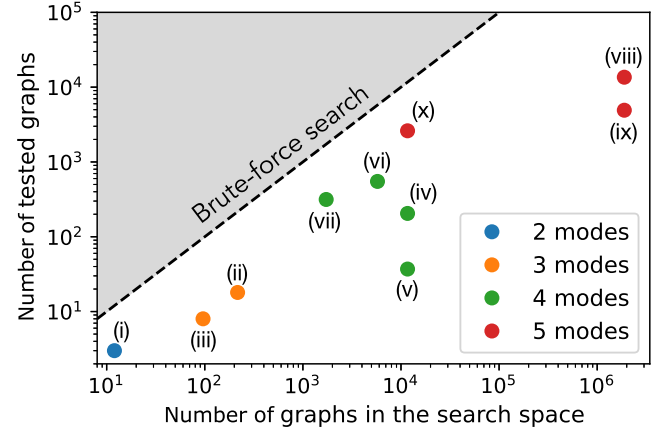


FIG. 7. Search-space reduction. We compare the number of graphs tested by AUTOSCATTER to the total number of graphs within the search space. A brute-force search would have to test every graph (dashed black line). Each data point corresponds to a different target functionality: (i) gyrator, (ii) isolator*, (iii) amplifier with intrinsic port losses*, (iv) three-port gyrator, (v) Wilkinson power divider, (vi) amplifier without intrinsic port losses*, (vii) circulator with only passive couplings between optical modes*, (viii) directional coupler*, (ix) resistive power divider, and (x) circulator without couplings between optical modes*. All examples marked with an asterisk (*) are discussed in Secs. IV–VII. See Supplemental Material [48] for a summary of further examples.

expect, the largest gains are obtained for complex devices featuring four or five modes because the brute-force search space grows exponentially with the squared number of modes in the device. It is not possible to predict, in general, how the performance gain evolves with mode number since that depends on the target device. In Supplemental Material [48], we provide more details on our analysis and discuss how the search-space structures of the individual target devices affect the respective performance gain. Furthermore, in Supplemental Material [48], we provide a summary of all target devices included in Fig. 7. With the help of the substantial performance gains displayed here, and for the mode numbers relevant for the variety of practically important target devices, our approach is able to provide the complete enumeration of solutions, which is one of its key advantages. It should be noted that eventually, for even further-increased mode numbers, one would have to resort to a combination of our search-space reduction with heuristic approaches.

IX. IMPLEMENTATION IN SPECIFIC EXPERIMENTAL PLATFORMS

In this section, we review the main steps to design a scattering device in a specific experimental platform. This section has two main motivations. The first motivation is to demonstrate how the ideal solutions discovered using AUTOSCATTER can be implemented in a wide range of

experimental platforms—including purely optical platforms, microwave circuits, optomechanics, and other hybrid platforms—covering a wide range of frequencies from MHz to hundreds of THz. In this framework, we show how to exploit the flexibility of our solutions to optimize secondary objectives, such as the device’s bandwidth or the number of active couplings. The second motivation to view our method in the broader context of the full experimental design pipeline is to highlight how it stands out as the only design step that is dependent on human ingenuity and leads to interpretable and transferable findings.

Typically, the first concrete choice in view of designing a scattering device is the type of port modes. This choice will depend on the functionality of the target device; e.g., for a microwave-to-optics converter, one needs a microwave and an optical mode with resonance frequency close to the respective target carrier frequencies.

Pairs of modes of the same type and with similar resonance frequency can naturally be coupled via a passive beam-splitter interaction [8,42], e.g., the evanescent coupling between two defect modes in a photonic crystal [8]. By itself, even a large difference between the resonance frequencies of two modes does not pose a fundamental constraint to their available linearized couplings because signals can be up- and down-converted by pumping a weak nonlinearity with a coherent drive. The specific coupling scheme depends on the experimental platform. In a purely optical setting, one can exploit the optical nonlinearities in the material, i.e., the Kerr or the $\chi^{(2)}$ nonlinearity. Mechanical modes and optical modes can be coupled via radiation pressure. In superconducting circuits, the coupling of microwave modes of different frequencies is based on the Josephson nonlinearity, driving the circuit with ac or a time-dependent magnetic flux piercing a superconducting quantum interference device (SQUID) [54]. While active couplings are widely available, some constraints to the quadratic Hamiltonian Eq. (1) and, thus, our graph connectivity might be present in specific experimental settings. For example, coupling a circuit microwave mode to an optical cavity mode typically requires a mechanical auxiliary mode [5].

The next and most important step in designing a new scattering experiment is to find a suitable RWA Hamiltonian that enforces the desired scattering behavior, taking into account possible hardware-specific constraints on the couplings. This step typically requires human ingenuity and is the central focus of many experimental and theoretical works [3,6,8,31,32,36,37,39,41,42]. Our method fully automates this task by finding all possible solutions and representing them using our graph language [see Fig. 1(a)].

At the level of the RWA, the solutions are highly transferable between hardware platforms, as one only needs to commit to a certain number of modes, coupling types, and the underlying cooperativities and synthetic field

fluxes, as we have shown in Appendix F. Since they do not depend on the resonance frequency of the modes, they leave open the choice of whether to implement certain couplings as active or passive couplings. For example, the isolator graph shown in Fig. 4(a) can be implemented fully actively, requiring three parametric pumps, as in Ref. [2]. Alternatively, it can be implemented using two parametric pumps and a passive coupling as in Ref. [42].

If one is willing to employ active couplings for every link, any desired graph can certainly be implemented. This approach, combining several pump tones at different frequencies, is common in different platforms [1–5,7,33,37,38,41]. However, implementing active couplings that rely on parametric driving may not be the most economical design in many scenarios. For example, in the optical regime, it is more challenging to combine many pump tones at different frequencies in this higher frequency range. For this reason, it is valuable to identify those setups that require the minimum amount of parametric pumps.

In many experimental platforms, one needs at least one active coupling to engineer a synthetic gauge flux [1–4,33,37,38,41], $\Phi_{i,j,l,\dots,k} = \arg(g_{ij}g_{jl}\dots g_{ki}) \neq 0$. Notable exceptions are setups featuring passive coupling between chiral modes [55,56] (see Supplemental Material [48]). Here, we exclude such exceptions and assume that all loops formed exclusively by passive couplings have zero flux. In this scenario, we can always choose a gauge in which all passive couplings are real valued. However, only beam-splitter interactions coupling modes with the same carrier frequency can be realized passively, which could be forbidden by other constraints in the graph. In particular, all complex-valued beam-splitter couplings and squeezing couplings are active and, thus, connect modes of different frequencies.

To determine the minimum number of active couplings, one has to solve a sorting problem, in which one sorts modes into frequency classes. If two modes are in the same class, they have the same carrier frequency; otherwise, they do not. The goal is to minimize the number of edges connecting modes of different classes under the aforementioned constraints. We solve this problem with a brute-force search (see Appendix I for more details). This search yields the minimum number of pumps as well as all possible combinations of passively coupled modes that allow us to reach the minimum. As an example, we analyze the number of active pumps for the fully directional coupler of Fig. 4(b). Our analysis shows that graph (I) requires at least four parametric pumps, while graphs (II) and (III) only require three pumps. Figure 8 displays two different optimal solutions for each graph. Graphs (II) and (III) only have these two solutions, while graph (I) has eight in total.

Even after choosing a graph and deciding which couplings are active and passive, our solutions still leave the freedom to choose the modes’ decay rates. We recall that the weights H_{ij} of our graphs are proportional to square roots of cooperativities, $H_{ij} = g_{ij}/\sqrt{\kappa_i\kappa_j}$ or $H_{ij} = \nu_{ij}/\sqrt{\kappa_i\kappa_j}$ [cf. Eq. (2)].

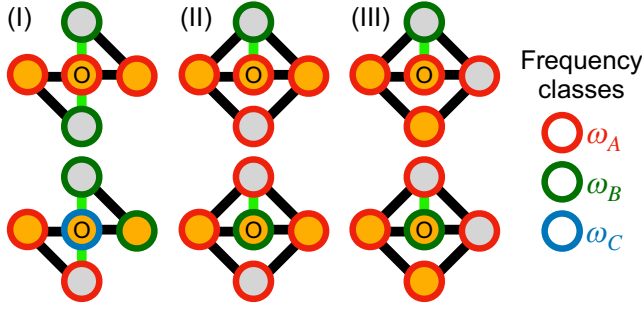


FIG. 8. Minimization of the number of active couplings. The graphs are annotated with the carrier frequency classes that minimize the number of active couplings for the directional coupler graphs from Fig. 4(b). Edges that connect two modes belonging to the same frequency class are coupled passively. The remaining couplings are active. A sorting is correct when it fulfills the constraint that complex-valued beam-splitter couplings (green) and squeezing (blue) interactions have to connect modes of different frequency classes. We find all possible sortings by using a brute-force search (see Appendix I for more details). The shown sortings minimize the number of active couplings. Graph (I) has six more such solutions (not shown).

Independent of the concrete values for the modes' decay rates κ_i , the scattering amplitudes at the carrier frequencies do not change as long as the coupling rates g_{ij} and ν_{ij} are rescaled accordingly. At the same time, the out-couplings and other loss rates can typically be tuned to a certain degree in experiments. This freedom can be used to change the frequency dependence of the scattering process, as illustrated in Fig. 9 for one of our directional couplers. By increasing the out-coupling rate of the input ports, one can increase the bandwidth of the transmission from the input to the output. With large losses on the auxiliary modes, one can suppress the reverse transmission over a broad frequency range.

Beyond the design choices discussed above, any hardware platform possesses its own hardware-specific challenges. The device should be engineered to host the desired high-quality-factor modes with the target passive quadratic couplings and strong enough high-order nonlinear interactions. This task requires domain-specific simulations of, e.g., the Maxwell equations or the elasticity equations. We note in passing that this task typically requires human ingenuity and, for this reason, has been the focus of intense efforts aimed at automatizing it (see, e.g., Refs. [57–59]). However, its platform-specific character makes any finding less transferable and generalizable. For this reason, it is outside of the scope of our work.

As a final step, the dynamics of the multimode system can be modeled using a set of coupled Langevin equations also featuring the nonlinear interactions, with bare couplings calculated from the domain-specific simulations. One can straightforwardly use this more general description to find the driving protocol to implement the target active couplings as detailed below. The pump frequencies ω_{ij}^P are

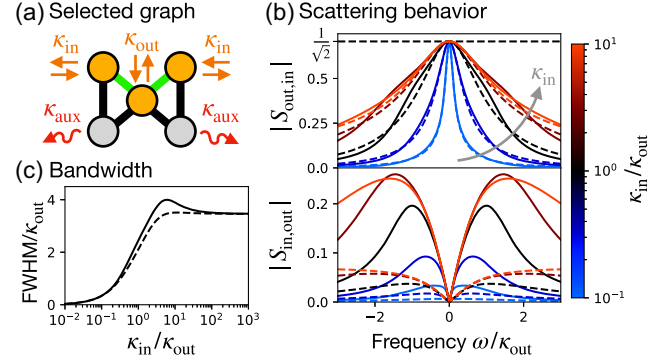


FIG. 9. Frequency-dependent scattering response. We show the scattering response of a directional coupler [graph (1) in Fig. 4(b)] for different values of the loss rates. (a) Graph representation labeled with the out-couplings (orange arrows) and intrinsic losses (red arrows). (b) Transmission from an input to the output port (top) and reverse transmission (bottom) as a function of the frequency (counted off from the carrier frequency) for different out-coupling ratios κ_{in}/κ_{out} (see color bar), and $\kappa_{aux}/\kappa_{out} = 2$ (solid lines) and 10 (dashed lines). The ideal transmission is $|S_{out,in}| = 1/\sqrt{2}$ (dashed black line) while the ideal reverse transmission $S_{in,out}$ is zero. Our approach optimizes the system, such that the coupler always displays the ideal behavior at the carrier frequency (for $\omega = 0$). (c) Bandwidth as a function of κ_{in} for $\kappa_{aux}/\kappa_{out}$ as in panel (b). We define the bandwidth as the full width at half maximum (FWHM) of the transmission $|S_{out,in}|$.

chosen to up and down-convert the signals to the target carrier frequencies $\omega_{L,j}$, e.g., for a nonlinearity based on three-wave mixing $\omega_{ij}^P = |\omega_{L,i} \pm \omega_{L,j}|$ (“+” for squeezing, “−” for beam-splitter couplings). Then, the steady-state amplitudes of the modes (also including pump modes that do not directly appear in the RWA description) are calculated from a set of nonlinear equations derived from the Langevin equations. By linearizing the Langevin equations around this steady state, one can express the quadratic couplings g_{ij} and ν_{ij} in terms of the bare nonlinear couplings and the steady amplitudes, and calculate target values for the latter quantities. In turn, the steady amplitudes can be adjusted by varying the powers and phases of the pumps. The appropriate values of the powers and phases can be estimated from the nonlinear equations for the steady state and can also be fine-tuned directly in the experiments. For most scattering devices, it is possible to achieve the ideal behavior without taking any asymptotic limit. In this scenario, the cooperativities are typically of order 1. Asymptotic solutions require large cooperativities. Such cooperativities can be reached in a variety of different hardware platforms featuring optical, microwave, and mechanical modes.

X. CONCLUSION

Our automated discovery algorithm AUTOSCATTER finds ideal high-level designs for real-world scattering devices

like transducers, amplifiers, isolators, or circulators. It provides interpretable and transferable solutions thanks to its graph representation, which encodes, in the same set of interpretable weights, a whole class of setups, unified by their scattering behavior but with different physical parameters. The flexibility in translating each solution back into physical parameters allows the user to implement them in a range of state-of-the-art experimental platforms and transfer them to different dissipative regimes, covering both Hamiltonian and reservoir engineering schemes. AUTOSCATTER can discover types of solutions that are normally difficult to obtain in an automated approach, such as asymptotic solutions. Through a series of examples, we have shown that the solutions generated by AUTOSCATTER lead to new insights, in the form of counter-intuitive design principles, generalizable design patterns, and analytical solutions to continuous classes of problems.

AUTOSCATTER is amenable to numerous future extensions, such as searching for optimized implementations in specific hardware platforms, neural-network-guided heuristics for larger devices where complete enumeration is infeasible, or targeting specific quantum noise characteristics. More generally, our results now open the door towards automated discovery in additional related domains of importance, like photonic transport through periodic structures composed of small modular blocks or scattering-based sensing experiments, as well as transfer to areas like electronic scattering transport. Nonlinear setups or even devices with functionalities based on time-dependent control require significant new developments but could still be treated with the same graph-based efficient search algorithm. Overall, the results shown here validate the fruitfulness of the main ideas guiding the field of artificial scientific discovery.

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DATA AVAILABILITY

The code used to produce the results of this manuscript is openly available in the GitHub repository AUTOSCATTER [9].

APPENDIX A: DETAILS OF THE DISCRETE OPTIMIZATION

During the optimization, we keep two libraries: one for valid graphs and one for invalid graphs. At the beginning of

the discrete optimization, our method tests only fully connected graphs for validity using the continuous optimization starting from the graph with zero auxiliary modes. It increments the number of auxiliary modes until it finds the minimum number required to achieve the target behavior, which is the smallest fully connected graph that satisfies the target behavior. Next, we investigate which couplings or coupling phases can be removed. This search space corresponds to a directed acyclic graph [see visualization in Fig. 2(b)], where the previously identified fully connected graph is the root node. To search through this space, we perform a breadth-first search, which means that our search always explores all nodes in the search space at the current depth before proceeding on to the next level starting from the root node.

In each iteration, we first check if the selected graph (the next node in the search space) is already in one of the two libraries. If so, the graph is skipped. Otherwise, it is tested by the continuous optimization. If no solution is found, the tested graph and all its subgraphs are added to the library of invalid graphs. Otherwise, the graph and all its extensions are added to the library of valid graphs. If some of the discovered parameters turn out to be zero, the corresponding simpler graph and all its extensions are added instead. The optimization is finished when all possible graphs have been sorted into one of the two libraries.

APPENDIX B: DETAILS OF THE CONTINUOUS OPTIMIZATION

For the minimization of the loss function in Eq. (3), we use the Broyden-Fletcher-Goldfarb-Shanno algorithm [60], an iterative gradient-descent method for solving nonlinear optimization problems. A valid solution is obtained whenever the loss reaches zero. The algorithm fails to find a solution if the loss becomes stuck in a nonzero minimum. Since the optimization landscape supports local minima, there is a risk of false-negative errors. To reduce this risk, we repeat the continuous optimization N_{rep} times. The hyperparameter N_{rep} is chosen empirically with a default value $N_{\text{rep}} = 10$.

During the discrete optimization, we assume that there always exists a valid graph. However, in our work, we identify two occasions where the loss function cannot be minimized to zero for any graph: (i) The desired target scattering behavior violates fundamental physics. For example, if you demand that the noise adopts a value below the quantum limit, no valid graph can be found. (ii) The desired scattering behavior can only be realized in a limiting case not covered by our optimization. If you ask our optimization scheme to search for the circulator behavior discussed in Fig. 6 without allowing for far-detuned modes, our continuous optimization will label any graph as invalid.

Please note that, in the main text, we have assumed that each port mode is coupled in reflection to a single waveguide. In the Supplemental Material [48], we discuss how

some alternative coupling schemes can be translated to our setups and how one can extend our approach to further coupling schemes, also covering setups including running-wave modes that are not invariant under time reversal, as in Ref. [44].

APPENDIX C: BDG HAMILTONIAN

We consider a hybrid system comprising N modes of different types and the most general quadratic time-independent Hamiltonian,

$$\hat{H} = \frac{1}{2} \sum_{j,k=1}^N (g_{jk} \hat{a}_j^\dagger \hat{a}_k + \nu_{jk} \hat{a}_j^\dagger \hat{a}_k^\dagger) + \text{H.c.} \quad (\text{C1})$$

Here, $\hat{a}_j$ and $\hat{a}_j^\dagger$ are the ladder operators of mode j , g is a Hermitian matrix, and ν can be chosen to be symmetric. For a more compact notation, it is convenient to group all the modes in a $2N$ -dimensional vector, $\hat{\xi} = \{\hat{a}_1, \dots, \hat{a}_N, \hat{a}_1^\dagger, \dots, \hat{a}_N^\dagger\}$, and to introduce the first-quantized BdG Hamiltonian

$$\hat{H} = \frac{1}{2} \hat{\xi}^\dagger H_{\text{BdG}} \hat{\xi}, \quad H_{\text{BdG}} = \begin{pmatrix} g & \nu \\ \nu^* & g^* \end{pmatrix}. \quad (\text{C2})$$

APPENDIX D: ROTATING FRAME

The Hamiltonian in Eq. (C1) is defined in a frame in which the phase space of each mode j rotates at the corresponding carrier frequency $\omega_{L,j}$. Thus, $-g_{jj}$ is interpreted as the detuning Δ_j of the carrier frequency $\omega_{L,j}$ from the mode resonance ω_j , $\Delta_j = \omega_{L,j} - \omega_j$. For multimode circuits, such a time-independent Hamiltonian is typically referred to as the rotating-wave Hamiltonian as it is obtained after a RWA that drops the fast-oscillating terms that result from nonresonant interactions. The corrections to the RWA approximation are small if all resonant frequencies ω_j are much larger than the coupling rates g_{ij} and ν_{ij} as well as the decay rates κ_j .

We note that if all couplings are active, the carrier frequencies $\omega_{L,j}$ for the different modes can be chosen independently of each other, adjusting the frequencies $\omega_{L,j}^P$ of the pump mediating the interaction such that $\omega_{L,j}^P = |\omega_{L,j} \pm \omega_j|$ (“+” for squeezing, “−” for beam-splitter couplings). Therefore, in experimental scenarios in which the mode resonances ω_j are not easily tunable, one can choose the carrier frequencies to implement the detunings Δ_j required in the scheme. On the other hand, two modes whose coupling is implemented passively (without a parametric pump) have the same carrier frequency; thus, one of the resonance frequencies, together with the common carrier frequency, has to be tuned to implement the target detunings.

Besides these constraints, the mode resonances ω_j are irrelevant in the high-level description based on the RWA. This appealing feature makes any insights gained using our method highly transferable across a variety of platforms. A more refined description including nonresonant interactions is typically used to quantify how a specific implementation deviates from the ideal target behavior. This deviation is device-specific and, thus, goes beyond the scope of our work.

APPENDIX E: LANGEVIN EQUATIONS

We use input-output theory to describe the dynamics of the open multimode circuit. As described in the main text, we distinguish between port modes, which are used as input and output ports for the signals, and auxiliary ports, which are not used as input ports and whose output is not monitored. Accordingly, all the losses in an auxiliary mode can be incorporated into a single loss channel. On the other hand, whenever we want to incorporate intrinsic losses in the port modes, we have to introduce two loss channels for these modes, an out-coupling and an intrinsic loss channel. Based on these considerations, we arrive at the Langevin equations

$$\begin{aligned} \dot{\hat{a}}_j(t) = & -i \sum_k g_{jk} \hat{a}_k(t) - i \sum_k \nu_{jk} \hat{a}_k^\dagger(t) - \frac{\kappa_j + \Gamma_j}{2} \hat{a}_j(t) \\ & - \sqrt{\kappa_j} \hat{a}_j^{\text{in}}(t) - \sqrt{\Gamma_j} \hat{a}_j^{\text{noise}}(t). \end{aligned} \quad (\text{E1})$$

Here, $\hat{a}_j^{\text{in}}$ is the input field for the main decay channel (for the port modes, the out-coupling channel), with decay rate κ_j . Likewise, $\hat{a}_j^{\text{noise}}$ is the noise entering port mode j from its intrinsic loss channel, which has decay rate Γ_j . The Langevin equations can be rewritten in a compact form as

$$\dot{\hat{\xi}}(t) = \left(-i \sigma_z H_{\text{BdG}} - \frac{\kappa + \Gamma}{2} \right) \hat{\xi}(t) - \sqrt{\kappa} \hat{\xi}^{\text{in}}(t) - \sqrt{\Gamma} \hat{\xi}^{\text{noise}}(t). \quad (\text{E2})$$

Here, we have grouped all input and noise fields in the vectors, $\hat{\xi}^{\text{in}} = \{\hat{a}_1^{\text{in}}, \dots, \hat{a}_N^{\text{in}}, \hat{a}_1^{\text{in}\dagger}, \dots, \hat{a}_N^{\text{in}\dagger}\}$ and $\hat{\xi}^{\text{noise}} = \{\hat{a}_1^{\text{noise}}, \dots, \hat{a}_N^{\text{noise}}, \hat{a}_1^{\text{noise}\dagger}, \dots, \hat{a}_N^{\text{noise}\dagger}\}$, respectively. Moreover, we have introduced the diagonal matrices $\kappa = \text{diag}(\kappa_1, \dots, \kappa_N, \kappa_1, \dots, \kappa_N)$, $\Gamma = \text{diag}(\Gamma_1, \dots, \Gamma_N, \Gamma_1, \dots, \Gamma_N)$ and

$$\sigma_z = \begin{pmatrix} \mathbb{1}_N & 0 \\ 0 & -\mathbb{1}_N \end{pmatrix}. \quad (\text{E3})$$

We note that the Langevin equations have the embedded particle-hole symmetry $\sigma_x \mathcal{K}$, where $\mathcal{K}$ denotes the complex conjugation and

$$\sigma_x = \begin{pmatrix} 0 & \mathbb{1}_N \\ \mathbb{1}_N & 0 \end{pmatrix}. \quad (\text{E4})$$

This finding simply reflects that the last N equations are the adjoint of the first N equations.

APPENDIX F: SCATTERING MATRIX

To fully characterize the linear response of a multimode circuit, one has to take into account that it depends on the frequency of the input field via the time derivative in the left-hand side of Eq. (E2). However, the ideal scattering behavior is only realized in a high-level description that focuses on signals that have a very smooth envelope and, thus, are spectrally well localized (in each port about the respective carrier frequency $\omega_{L,j}$). Taking into account that in the rotating frame the frequency is counted off from the respective carrier frequency $\omega_{L,j}$, the response of interest is the “zero-frequency” response obtained by setting $\hat{\xi}(t) = 0$ in Eq. (E2). After calculating $\hat{\xi}$ in this way and plugging it into the input-output relations

$$\hat{\xi}^{\text{out}} = \hat{\xi}^{\text{in}} + \sqrt{\kappa} \hat{\xi}, \quad (\text{F1})$$

we find

$$\hat{\xi}^{\text{out}} = S \hat{\xi}^{\text{in}} + \mathcal{N} \hat{\xi}^{\text{noise}} \quad (\text{F2})$$

with

$$S = \mathbb{1}_{2N} + \left(-i\sigma_z H - \frac{\gamma}{2} - \frac{\mathbb{1}_{2N}}{2} \right)^{-1}, \quad (\text{F3a})$$

$$\mathcal{N} = (S - \mathbb{1}_{2N}) \sqrt{\gamma}. \quad (\text{F3b})$$

Here, S is the scattering matrix, and the matrix $\mathcal{N}$ describes the linear response to the fluctuations entering the circuit from the intrinsic loss channels. In view of discovering classes of solutions displaying the same scattering behavior, we find that it is possible to eliminate the explicit dependence of S and $\mathcal{N}$ on the decay rates κ_i by appropriately rescaling the BdG Hamiltonian and the intrinsic losses by introducing the rescaled Bogoliubov Hamiltonian H as

$$H = \frac{1}{\sqrt{\kappa}} H_{\text{BdG}} \frac{1}{\sqrt{\kappa}}, \quad \gamma = \Gamma \kappa^{-1}. \quad (\text{F4})$$

We note that H is parametrized by N rescaled detunings $H_{ii} = -\Delta_i/\kappa_i$ with $1 \leq i \leq N$, $N(N-1)/2$ dimensionless couplings $g_{ij}/\sqrt{\kappa_j \kappa_i}$ with $1 \leq i < j \leq N$, and $N(N+1)/2$ squeezing amplitudes $\nu_{ij}/\sqrt{\kappa_j \kappa_i}$ with $1 \leq i \leq j \leq N$. These parameters are learning parameters in the continuous optimization. As pointed out in the main text, the functionality of the device depends only on the dimensionless detunings H_{ii} , the cooperativities $C_{ij} = 4|H_{ij}|^2$, and the gauge-invariant

geometrical phases $\Phi_{i,j,l,\dots,k} = \arg(H_{ij}H_{jl}\dots H_{ki})$ accumulated in closed loops (synthetic field fluxes) [see Eq. (F3a)]. If we allow for intrinsic loss channels in the port modes, the behavior is furthermore influenced by the dimensionless intrinsic loss rates γ [see Eqs. (F3a) and (F3b)].

We note that, in the most general case, the S matrix is a $2N \times 2N$ matrix [cf. Eqs. (C2), (F3a), and (F4)]. This dimensionality reflects the fact that the intensity of the transmitted field can depend on the phase of the input field.

To simplify the discussion in the main text, we have focused on phase-preserving devices. For such devices, the scattering matrix as defined in Eq. (F3a) has a block-diagonal form with two $N \times N$ blocks mapped one into the other by particle-hole symmetry [39]. This form allows one to define the scattering matrix S as one of the two blocks obtained from Eq. (F3a) and, thus, to arrive at an $N \times N$ matrix, consistent with Eq. (3). In this case, our program does not calculate the “large” $2N \times 2N$ matrix but rather only one block.

For a purely routing device (without a squeezing interaction, $\nu = 0$), the diagonal blocks correspond to the annihilation and creation operators, respectively. Thus, the “smaller” scattering matrix can be taken to be the block $S_{1:N,1:N}$ of the “larger matrix” per Eq. (F2), which is also the standard definition of the scattering matrix for excitation conserving systems.

For phase-preserving amplifiers, we enforce the block structure of the “large” scattering matrix by constraining the matrix H in the following way: We divide the modes into two subsets M_1 and M_2 with N_1 and $N_2 = N - N_1$ modes, respectively. Then, we couple modes in the same set exclusively with beam-splitter interactions and modes in different sets via two-mode squeezing interactions. Consequently, the set of equations of motion for $\hat{\xi} = \{\hat{a}_1, \dots, \hat{a}_N, \hat{a}_1^\dagger, \dots, \hat{a}_N^\dagger\}$ [see Eq. (E1)] breaks down into two decoupled sets of equations for $\hat{\xi}$ and $\hat{\xi}^\dagger$, which are related to each other via particle-hole symmetry. Note that $\hat{\xi}$ contains all $\hat{a}_j$ with $j \in M_1$ and all $\hat{a}_j^\dagger$ with $j \in M_2$ and decouples from $\hat{\xi}^\dagger$, which contains the respective conjugate operators. Both decoupled sets correspond to the same graph. Our optimization automatically tests all possible assignments of the modes into the two subsets M_1 and M_2 .

Given a graph, one can read out which modes are in which subset in the following way: Modes coupled via a beam-splitter interaction are within the same subset; modes coupled via a squeezing interaction are in different subsets. By applying this rule step by step for each coupling, one has quickly characterized all modes.

The resulting scattering matrix has a block structure because the output fields $a_{i \in M_1}^{\text{out}}$ and $a_{i \in M_2}^{\text{out}\dagger}$ depend only on the input fields $a_{i \in M_1}^{\text{in}}$ and $a_{i \in M_2}^{\text{in}\dagger}$, and, likewise, for the fields $a_{i \in M_2}^{\text{out}}$, $a_{i \in M_1}^{\text{out}\dagger}$, $a_{i \in M_2}^{\text{in}}$, and $a_{i \in M_1}^{\text{in}\dagger}$. This structure allows us to take as a scattering matrix the block encoding the response of the output fields $a_{i \in M_1}^{\text{out}}$ and $a_{i \in M_2}^{\text{out}\dagger}$ to the

input fields $a_{i \in M_1}^{\text{in}}$ and $a_{i \in M_2}^{\text{in}\dagger}$. In practice, we calculate the “small” $N \times N$ scattering matrix by replacing all the $2N \times 2N$ matrices ($\mathbb{1}_{2N}$, σ_z , H , and γ) in Eq. (F3a) with the relevant $N \times N$ block; e.g., we replace σ_z with

$$\begin{pmatrix} \mathbb{1}_{N_1} & 0 \\ 0 & -\mathbb{1}_{N_2} \end{pmatrix}. \quad (\text{F5})$$

APPENDIX G: STABILITY

In the presence of squeezing interactions, the Langevin equations in Eq. (E1) can describe unstable motions drifting away from a saddle point. We can rewrite Eq. (E2) in the form

$$\dot{\hat{\xi}}(t) = D\hat{\xi}(t) - \sqrt{\kappa}\hat{\xi}^{\text{in}}(t) - \sqrt{\Gamma}\hat{\xi}^{\text{noise}}(t), \quad (\text{G1})$$

with $D = -i\sigma_z H_{\text{BdG}} - (\kappa + \Gamma)/2$ as the dynamical matrix of the coupled system. The motion is stable as long as the dynamical matrix D has only eigenvalues with a negative real part.

Since the solutions obtained using our method are not necessarily stable, we exclude unstable solutions after checking their stability. We note that the eigenvalues of D , as well as the signs of their real parts, can depend on the actual choice of κ_j , which can lead to constraints on these parameters.

APPENDIX H: NOISE ANALYSIS

The field operators for the input and noise channels $\hat{a}_j^{\text{in}}$ and $\hat{a}_j^{\text{noise}}$ obey the correlators

$$\langle \hat{a}_j^{\dagger,s}(t) \hat{a}_j^s(t') \rangle = n_j^s \delta(t - t'), \quad (\text{H1a})$$

$$\langle \hat{a}_j^s(t) \hat{a}_j^{\dagger,s}(t') \rangle = (n_j^s + 1) \delta(t - t'), \quad (\text{H1b})$$

with s standing either for “in” or “noise.” Here, n_j^{in} and n_j^{noise} are the occupation numbers of the baths connected to the respective mode.

For the phase-preserving amplifier discussed in Fig. 4(c), the number of added noise photons on the input channel (mode 1) and output channel (mode 2) equals

$$N_{\text{input}}^{\text{add}} = \frac{1}{2} \sum_k ((n_k^{\text{in}} + 1) |S_{1k}|^2 + (n_k^{\text{noise}} + 1) |\mathcal{N}_{1k}|^2), \quad (\text{H2a})$$

$$N_{\text{output}}^{\text{add}} = \frac{1}{2} \sum_{k \neq 1} ((n_k^{\text{in}} + 1) |S_{2k}|^2 + (n_k^{\text{noise}} + 1) |\mathcal{N}_{2k}|^2). \quad (\text{H2b})$$

The quantum limits for a phase-preserving amplifier are $N_{\text{input}}^{\text{add}} = 1/2$ and $N_{\text{output}}^{\text{add}} = (G - 1)/2$ [40]. If the two modes

are in different subsets M_1 and M_2 (see Appendix F), the added noise on the output is increased by one photon, meaning that the minimum number of added photons equals $N_{\text{output}}^{\text{add}} = (G + 1)/2$ in this case.

We assume that all the baths are at zero temperature, except the bath connected to the output channel. Thus, $n_j^{\text{in}} = 0$ if $j \neq 2$, $n_j^{\text{noise}} = 0$ for all j , and $n_2^{\text{in}} > 0$.

During optimization, we enforce that the noise on the input and output channels equals their respective quantum limits by adding the two additional terms $|f_1|^2 = (N_{\text{input}}^{\text{add}} - 1/2)^2$ and $|f_2|^2 = (N_{\text{output}}^{\text{add}} - (G \pm 1)/2)^2$ to the cost function Eq. (3). Therefore, all identified architectures automatically fulfill these equality constraints. The sign in constraint f_2 is chosen according to the tested graph. If input and output modes are in the same subset M_1 or M_2 , we choose the minus sign. Otherwise, we choose the plus sign.

APPENDIX I: MINIMAL NUMBER OF PARAMETRIC INTERACTIONS

Here, we summarize our algorithm for finding the minimum number of parametric pumps needed to realize a given graph. The key problem is to sort the graph’s nodes into frequency classes. If two modes are in the same class, they have the same frequency; otherwise, they do not. Each graph edge connecting modes of two different classes requires a pump. The goal is to minimize the number of such edges under the constraint that the modes coupled by squeezing (blue) or complex beam-splitter interactions (green) must be in different classes.

We solve this problem in the following way: Iterating through all nodes, we assign to the current node either one of the frequency classes already assigned to a previous node, or we assign a new one, which was not used before. After all nodes have been labeled, we test whether modes coupled by squeezing (blue) or complex beam-splitter interactions (green) are in different classes. If so, we count the number of pumps. Otherwise, this labeling is skipped. We repeat this algorithm until every possible distribution of frequency classes has been tested. This search yields the minimum number of pumps as well as all possible combinations of carrier frequencies that must be chosen equal to each other so that this minimum is reached.

The task described above is in fact a so-called “graph coloring” problem. In our setting, the frequency class plays the role of the “color.”

APPENDIX J: GAUGE FREEDOM AND GAUGE FIXING

The scattering matrix is a complex matrix that encodes the amplitudes and phases of the output fields in response to a set of input fields. Many experiments focus on the amplitude of the fields. Nevertheless, the phases can also be observed after fixing a gauge by committing to specific

experimental settings and physical interpretations of the fields $\hat{a}_j^{\text{in/out}}$. However, we do not want to distinguish between phases that depend on the detailed arrangement of the experimental setup or equivalent physical interpretations. For this purpose, we allow for the free parameters $\gamma_j^{\text{in/out}}$ in our cost function Eq. (3). The goal of this appendix is to make an explicit connection between this gauge freedom and the aforementioned experimental settings and physical interpretations.

We start by discussing the freedom of choosing the reference position where the input and output fields are evaluated. Assume that we change the reference position for port j and that the new reference position is further away from the device. An incoming signal acquires the additional phase ϕ_j while traveling from the new reference point to the old reference point. The same phase will also be acquired by an outgoing signal when traveling from the old to the new reference point. In other words, the transfer matrix will change according to the rule $S_{jk} \rightarrow S_{jk} e^{i(\phi_j + \phi_k)}$, where $\phi_j + \phi_k$ is the overall additional phase acquired by a signal traveling from the new reference position in port k to the new reference position in port j . Comparing to Eq. (3), we see that $\gamma_j^{\text{out}} = -\gamma_j^{\text{in}} = \phi_j$.

An additional gauge freedom for the fields $\hat{a}_j^{\text{in/out}}$ comes from the arbitrary phase ϕ_j in the transformation to the rotating frame, $\hat{a}_j^{\text{in/out}} = \exp[i(\omega_{L,j}t + \phi_j)]\hat{a}_{\text{lab},j}^{\text{in/out}}$, where $\hat{a}_{\text{lab},j}^{\text{in/out}}$ are the fields in the lab frame. The phase ϕ_j determines the physical interpretation of the quadratures $\hat{X}_j^{\text{in/out}}$ and $\hat{P}_j^{\text{in/out}}$ (real and imaginary parts of the field),

$$\hat{X}_j^{\text{in}} = \hat{X}_{\text{lab},j}^{\text{in}} \cos(\omega_{L,j}t + \phi_j) - \hat{P}_{\text{lab},j}^{\text{in}} \sin(\omega_{L,j}t + \phi_j). \quad (\text{J1})$$

The quadratures of the field in the lab frame have the same physical interpretation at all ports; e.g., in electromagnetism, $\hat{X}_{\text{lab},j}^{\text{in}}$ could correspond to the amplitude of an electric field with a fixed polarization and $\hat{P}_{\text{lab},j}^{\text{in}}$ to the amplitude of the transverse magnetic field. For incommensurate carrier frequencies $\omega_{L,j}$, any special initial choice of phases ϕ_j , e.g., $\phi_j = \phi$, would quickly dephase. As a consequence, all gauges ϕ_j are physically equivalent. To set them on equal footing, we allow for arbitrary $\gamma_j^{\text{out}} = \gamma_j^{\text{in}} = \phi_j$ in Eq. (3). Combining this gauge freedom with the choice of a reference position in each port waveguide, we arrive at independent values for γ_j^{out} and γ_j^{in} in Eq. (3).

In the special case of equal carrier frequency $\omega_{L,j} = \omega_L$, the uniform gauge $\phi_j = \phi$ has a special physical meaning. In this gauge, the physical interpretation of the quadratures depends on time but remains the same at every port [see Eq. (J1)]. In this experimental scenario, the target scattering matrix is usually written in the uniform gauge, and one should restrict the gauge freedom to $\gamma_j^{\text{out}} = -\gamma_j^{\text{in}} = \phi_j$.

To highlight the importance of the right choice for the gauge freedom, we consider a concrete example in the gyrator (see Supplemental Material [48]). The gyrator is a nonreciprocal device with scattering matrix

$$S_{\text{target}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (\text{J2})$$

For a gyrator, it is assumed that the carrier frequencies and the physical meaning of the fields at the two ports are the same. With this assumption, we can interpret the scattering matrix as describing a nonreciprocal device in which a field injected into the first port exits the second port as the same field, e.g., an electric field, but with an opposite sign. On the other hand, exchanging the emitter and the receiver, a field entering from the second port exits the first port without any sign change.

On the contrary, without the assumption that the quadratures have the same physical interpretation at both ports, the sign difference between the two off-diagonal elements loses any meaning. Indeed, it can be gauged away with the appropriate choice of γ_j^{out} and γ_j^{in} in Eq. (3), e.g., $\gamma_2^{\text{out}} = \pi$ and $\gamma_2^{\text{in}} = \gamma_1^{\text{in/out}} = 0$.

As the choice of gauge freedom is platform dependent, AUTOSCATTER allows the user to choose between four different options for handling the gauge freedoms: (i) no gauge freedom, (ii) allow only for free reference positions of the ports ($\gamma_j^{\text{out}} = -\gamma_j^{\text{in}}$), (iii) allow only for the gauge freedoms of the fields $\hat{a}_j^{\text{in/out}}$ ($\gamma_j^{\text{out}} = \gamma_j^{\text{in}}$), and (iv) allow for both degrees of freedom (independent γ_j^{out} and γ_j^{in}).

In all examples presented in the main text, one could use both options (ii) and (iv) to arrive at the same results. However, as we discussed using the gyrator as an example, option (ii) is sometimes implicitly assumed for certain nonreciprocal devices. For this reason, we have used this option to produce the data in Fig. 7 and Supplemental Material [48].

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